

Carnegie Mellon University
The Robotics Institute



Recording Reminder

- ☐ This session will be audio recorded for educational use by other students in this course.

Project Logistics

- ❑ Project presentations: Dec 3 (Tue) and 5 (Thu)
- ❑ 15 teams on Tue, 14 teams on Thu. Check the schedule on Piazza!
- ❑ Grading: 20% pre + 20% final report

Project presentation logistics:

Each team will have **4 minutes** to present, followed by a **1-minute** Q&A session. The presentation will be cut off at 4 minutes, so **please practice well**. It is up to you whether one person presents or everyone presents (encouraged). We also encourage everyone to participate in the project presentation. Knowing what other people are working on is an extremely important part of robot learning research.

Project presentation slides:

We will use one computer to present all the slides. For Tue groups, please post your slides in the correct order [here](#). For Thu groups, [here](#). 4 minutes are short, so organize your slides strategically. Please check out the first page in the shared Google slides for guidance. *Important note:* Don't forget to **grant read permissions to everyone** to play videos during the presentation. This is a very common mistake, and we won't be able to grant you extra time if you forget to do so.

Project Logistics

- ❑ Project presentations: Dec 3 (Tue) and 5 (Thu)
- ❑ 15 teams on Tue, 14 teams on Thu. Check the schedule on Piazza!
- ❑ Grading: 20% pre + 20% final report

Project final report:

Due on Dec 13. You can NOT use the late days for the final report. Only one submission is needed per group. The page limit is 8 (including references), and please use the [CoRL template](#). Make sure to cover the following questions:

- Motivation: Why is it important and relevant? Why should we care?
- Problem statement & prior work: Define the research problem. Briefly mention related works that are relevant to your idea instead of covering all related works.
- Your idea: What is the (key) idea?
- Your idea: Why does your idea make sense intuitively?
- Your idea: How does it relate to prior work in the area?
- Your results: If it worked, how much did you improve and why?
- Your results: If it did not work, why did you expect it to work? What are the next steps to try?
- Your results: Any negative results? Maybe you figured after trying that it did not make sense.
- Summary
- Future directions

Recap: Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Today's Agenda

Today is going to be a “causal” lecture! Let's make it interactive so raise your hand for any questions at anytime.

- ☐ Rethink: What is robot learning and why is it so hard?
- ☐ (Briefly) review what we have learned
- ☐ Open challenges and discussions
- ☐ Q&A! Ask me anything!

Today's Agenda

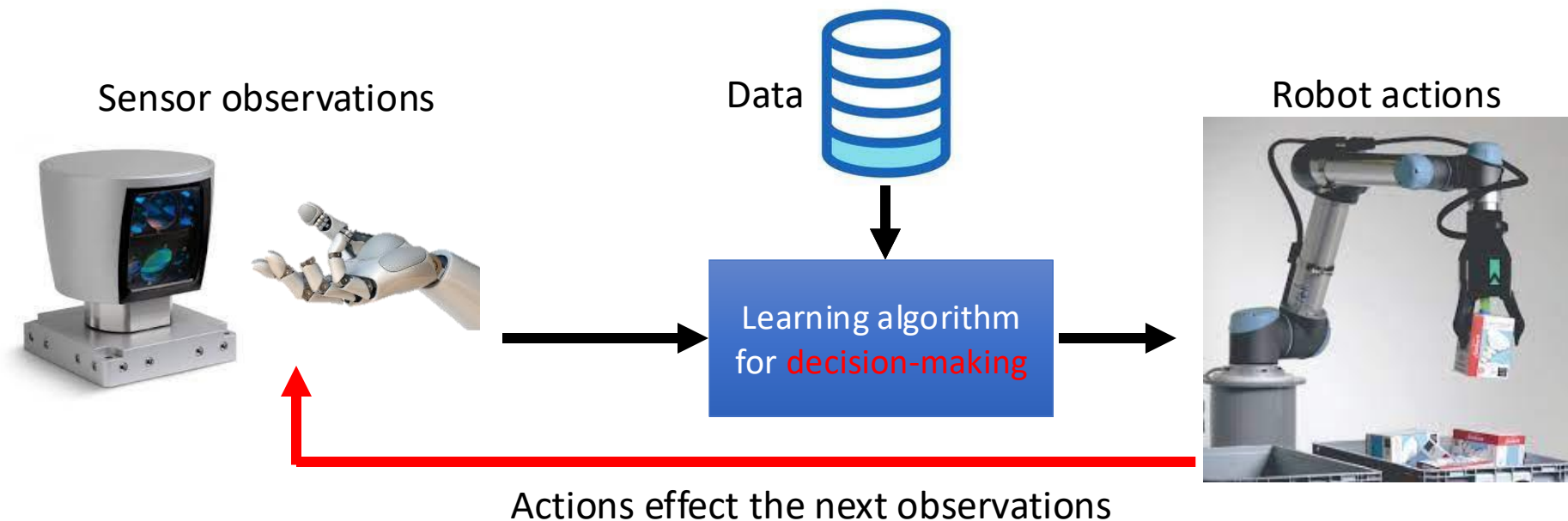
Today is going to be a “causal” lecture! Let's make it interactive so raise your hand for any questions at anytime.

- ☒ Rethink: What is robot learning and why is it so hard?
- ☐ (Briefly) review what we have learned
- ☐ Open challenges and discussions
- ☐ Q&A! Ask me anything!

What is “Robot Learning”?

What is “robot learning” and what is this class about?

❑ **Learning** to make **sequential** decisions in the **physical world**

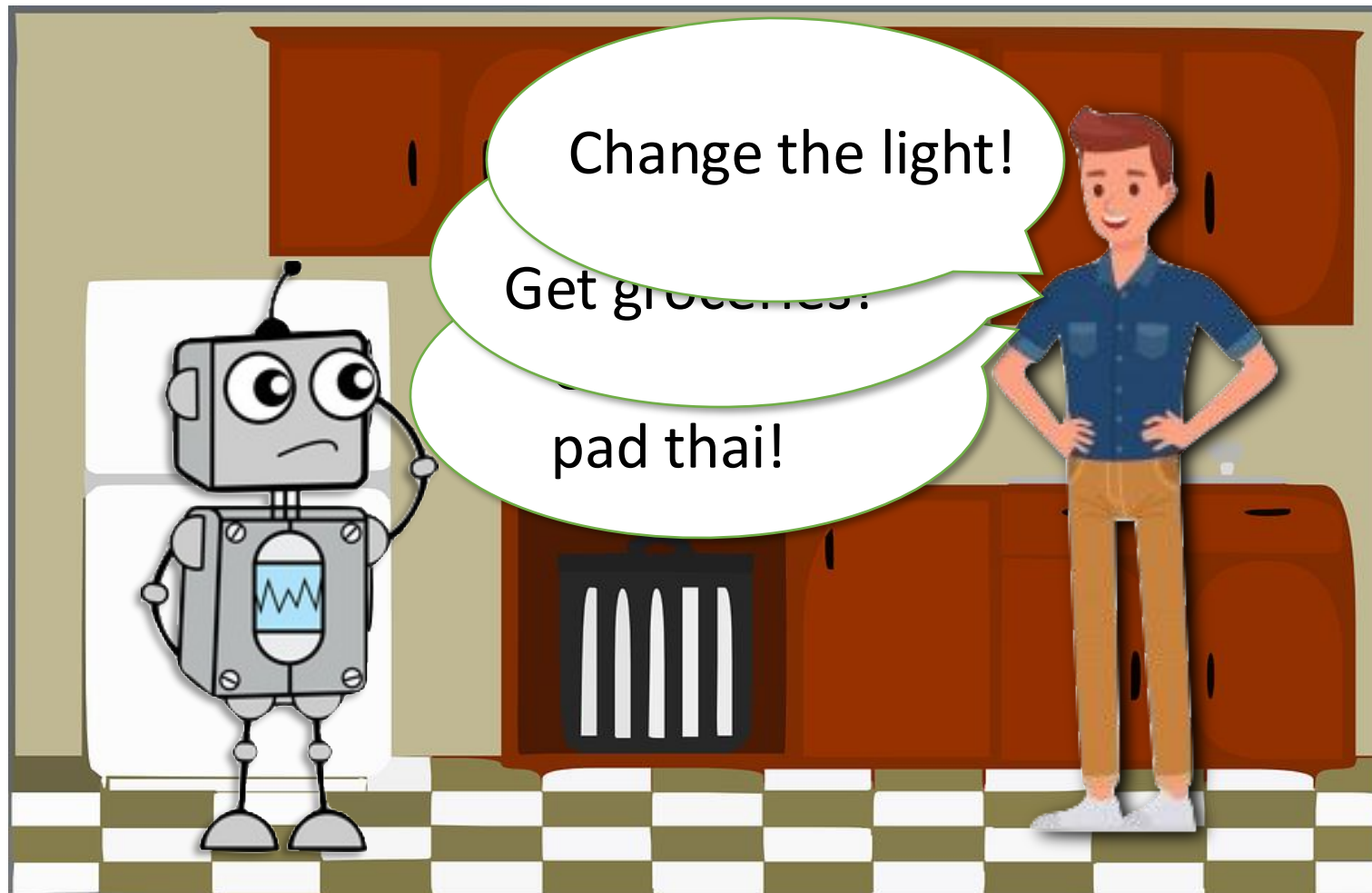


What is “Robot Learning”?

- ❑ **Learning** to make **sequential** decisions in the **physical world**
- ❑ **Learning**: Data-driven and improve from data
 - W/o learning & data: search & planning, classic control, optimal control, ...
- ❑ **Sequential**: The current action/decision influences the next state therefore the next action/decision
 - W/o sequential: bandit, standard supervised learning, ...
- ❑ **Physical world**: The robot needs to interact with the physical world in the closed-loop
 - A.k.a. “embodied intelligence”
 - W/o physical world: RL for games, LLMs...

What is the Goal of “Robot Learning”?

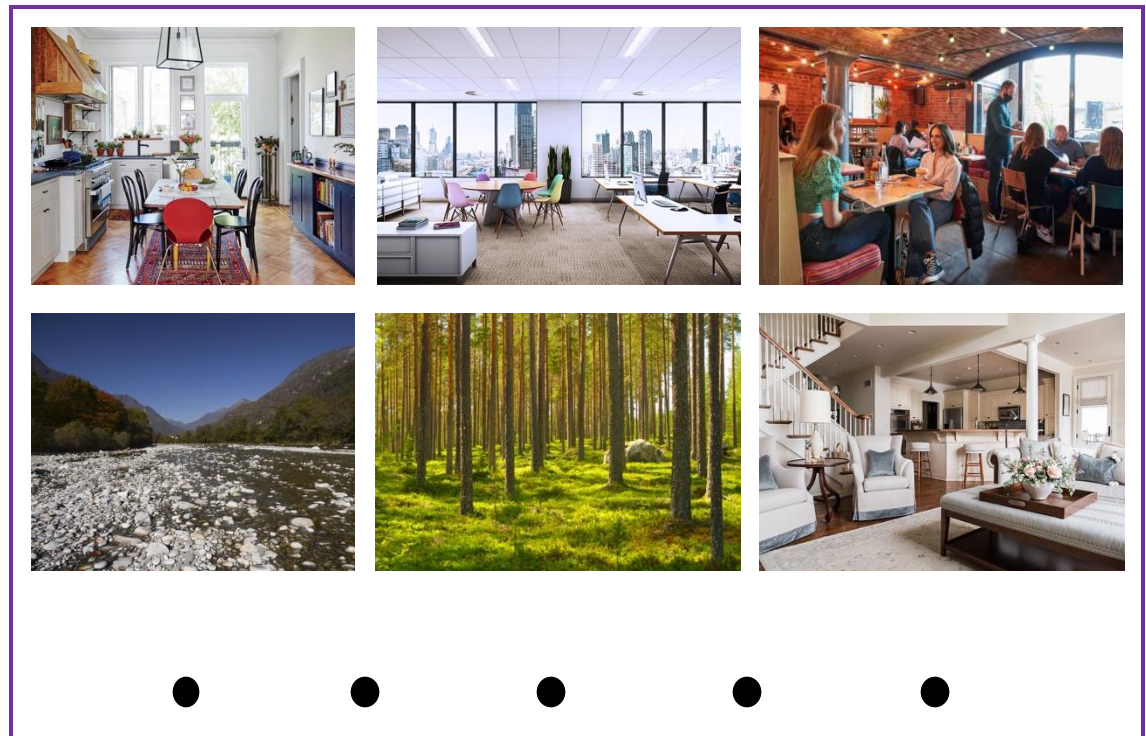
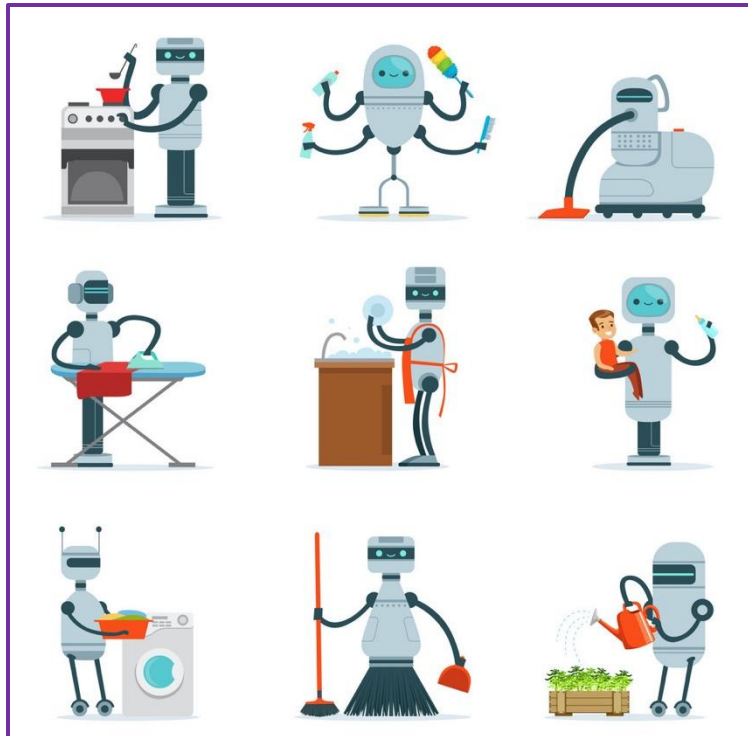
- ❑ Ultimate goal: Build **general-purpose embodied** intelligence by learning to make sequential decisions in the physical world



What is the Goal of “Robot Learning”?

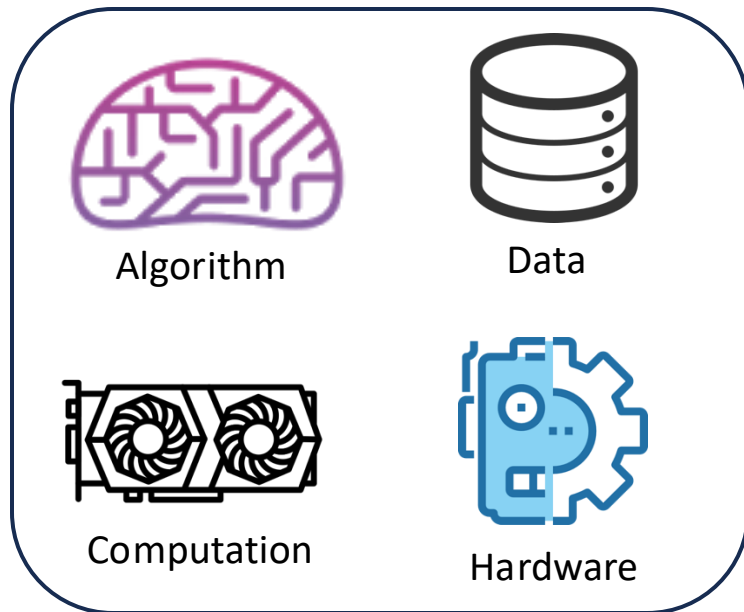
- ❑ Ultimate goal: Build **general-purpose embodied** intelligence by learning to make sequential decisions in the physical world

Robots that can do thousands of tasks in thousands of environments



What is the Goal of “Robot Learning”?

- ❑ Ultimate goal: Build **general-purpose embodied** intelligence by learning to make sequential decisions in the physical world



How far are we?



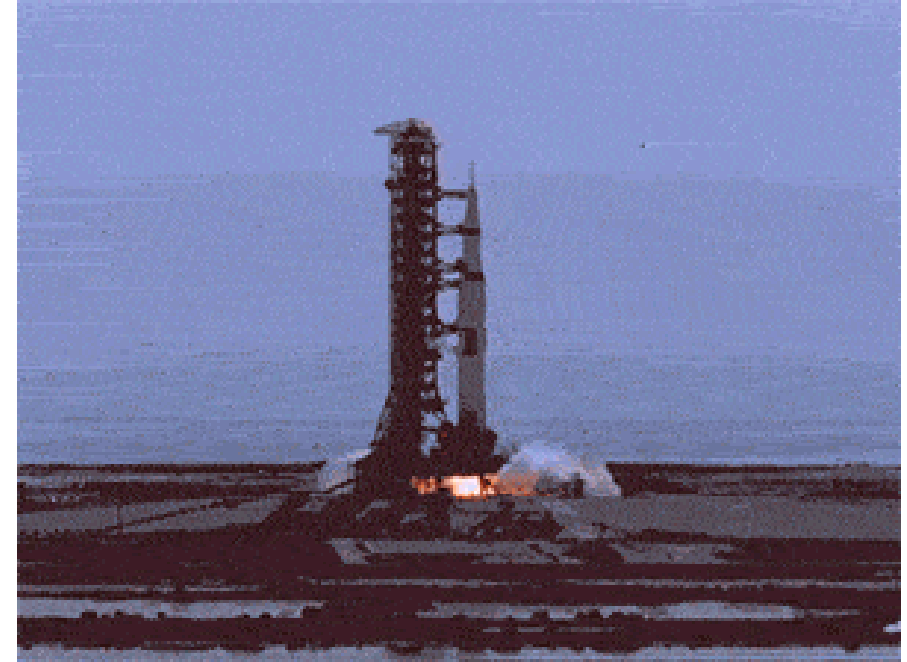
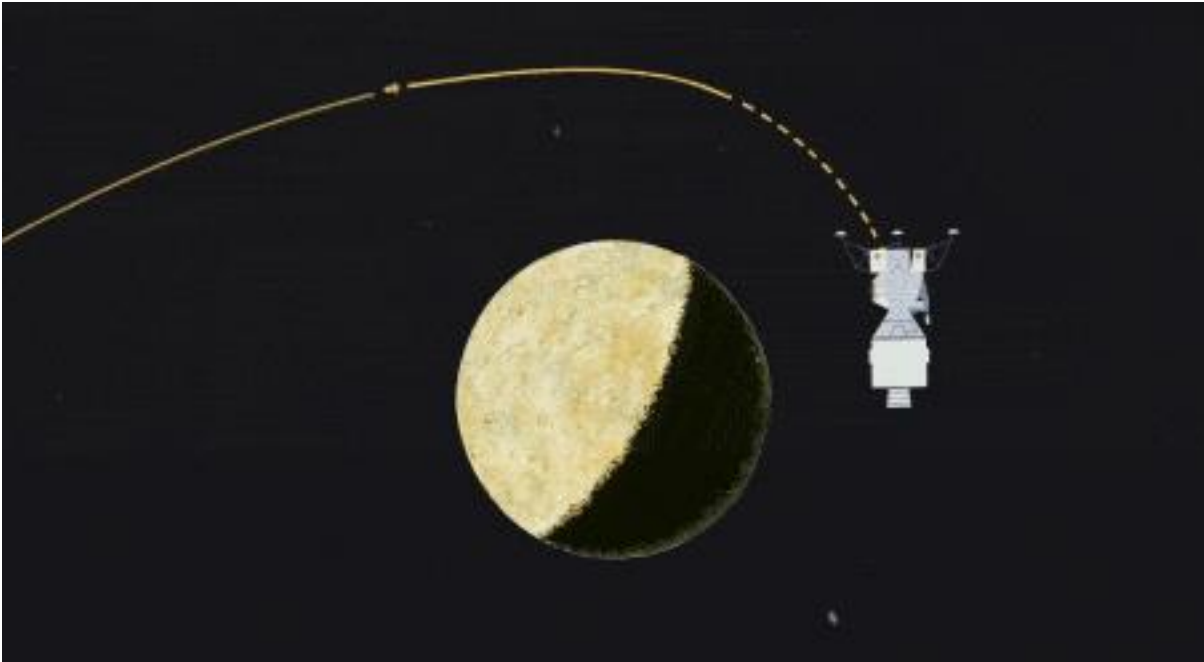
Embodied AGI in the real physical world



- ❑ We’ve made (a lot of) progress in terms of **domain-specific embodied intelligence**, either learning-based or not
- ❑ However, we are still very far from general-purpose embodied intelligence!

Where are We Today: Non-learning Method

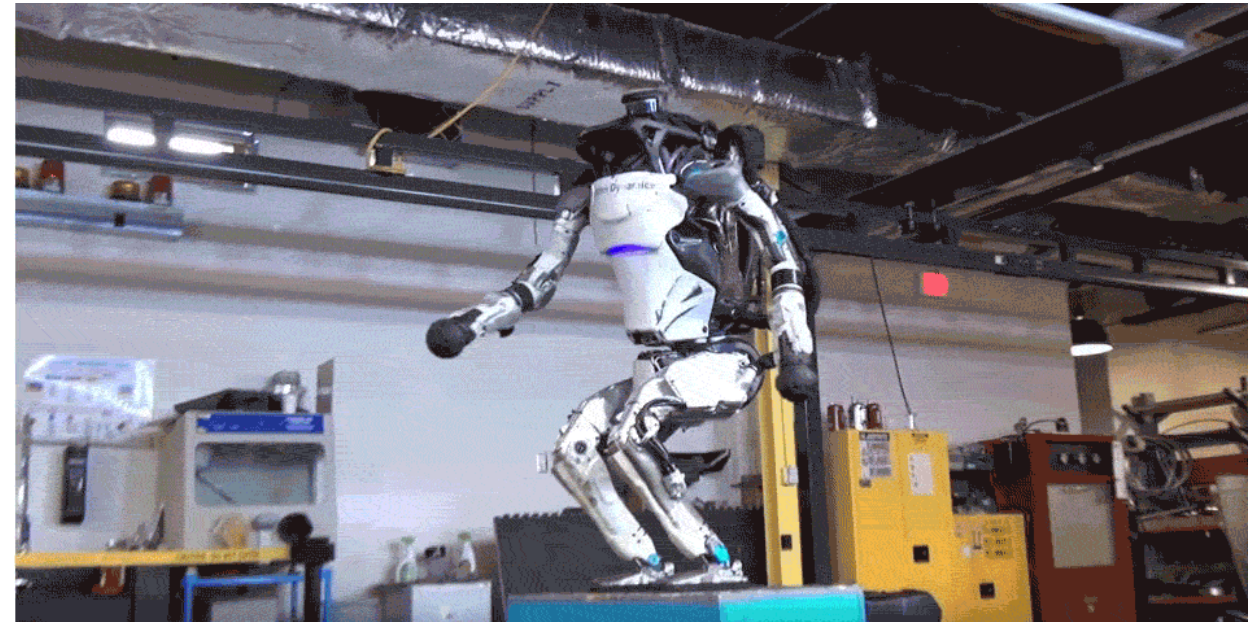
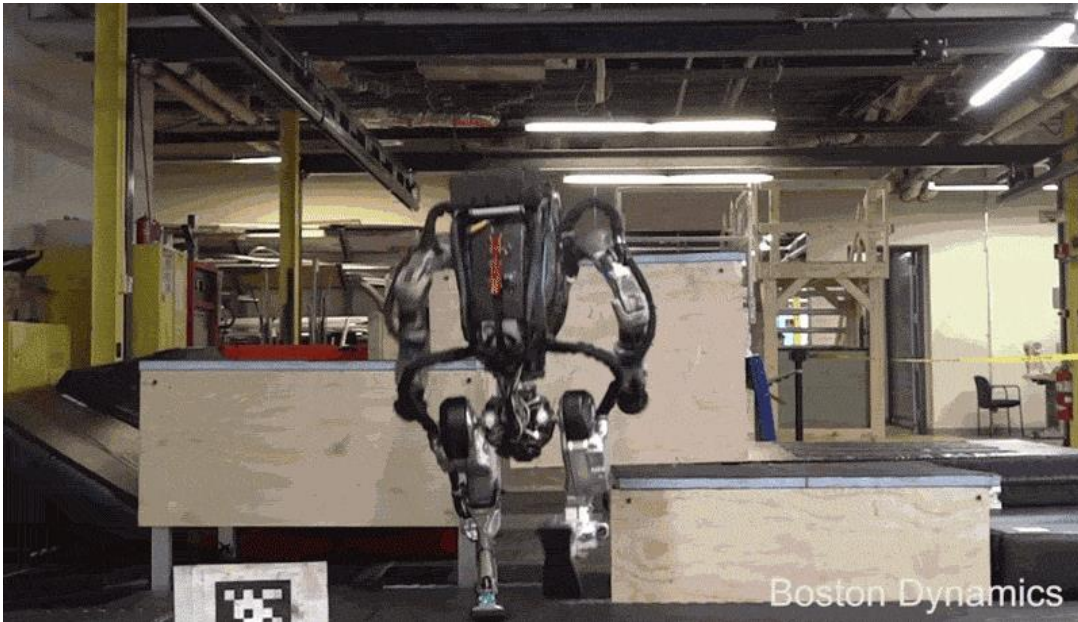
- ❑ Before seeing today's progress in robot learning, let's first see some results w/o learning:



Apollo 11 trajectory optimization and control: optimal control + robust control

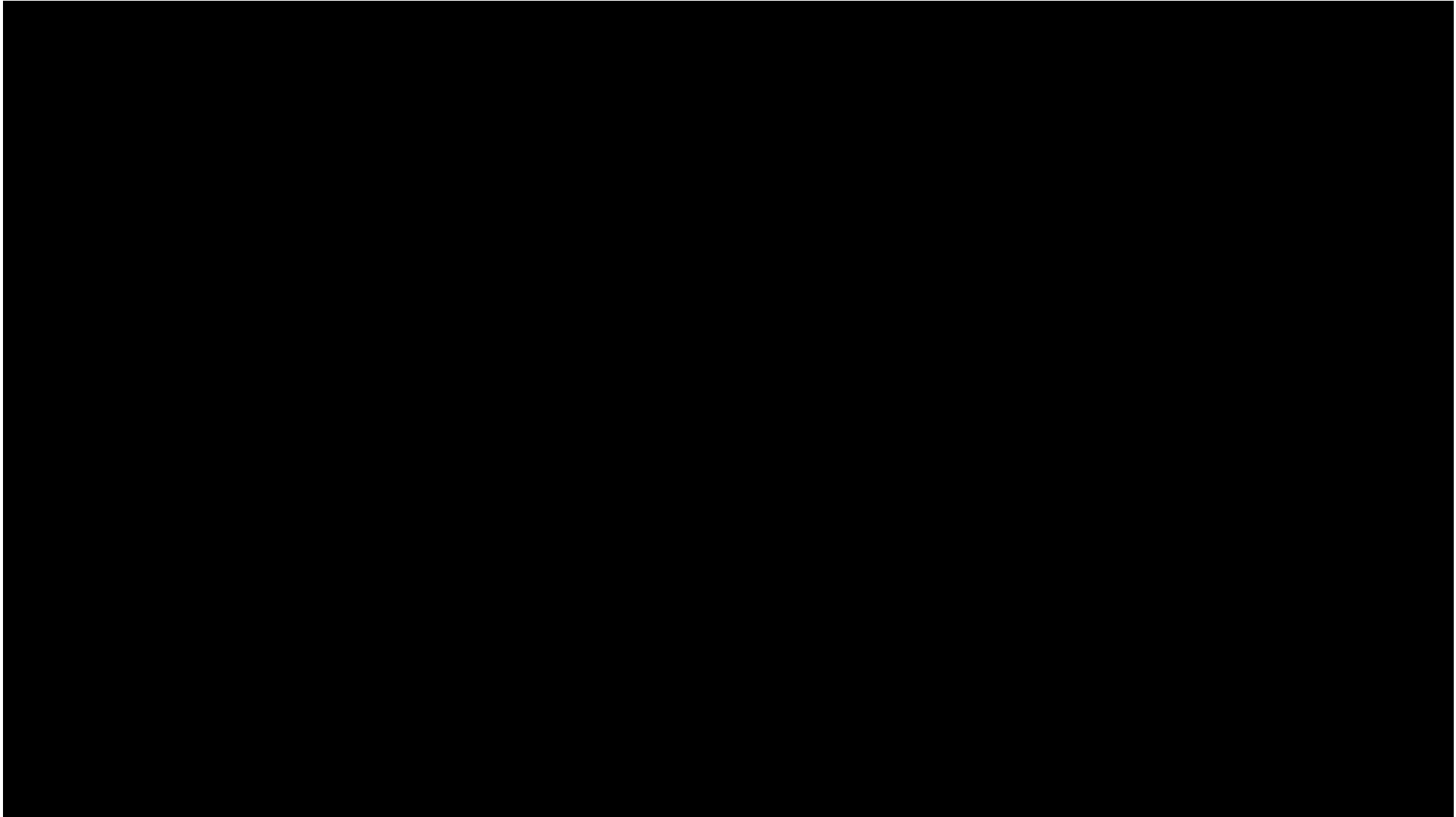
Where are We Today: Non-learning Method

- ❑ W/o learning:



Boston Dynamics Atlas humanoid planning and control: trajectory optimization + MPC

Where are We Today: Non-learning Method



Farewell to Atlas: Boston Dynamics Atlas officially retired this year (April 16)

Where are We Today: Non-learning Method

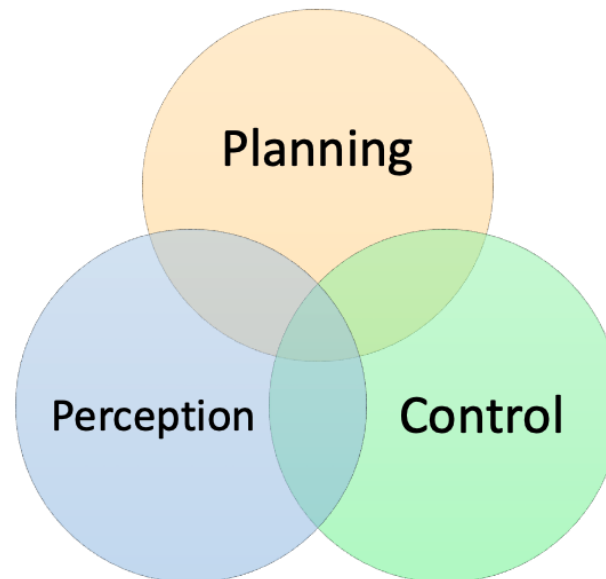
- ❑ W/o learning:



High-speed robot hand (Ishikawa Komuro Lab) in 2009

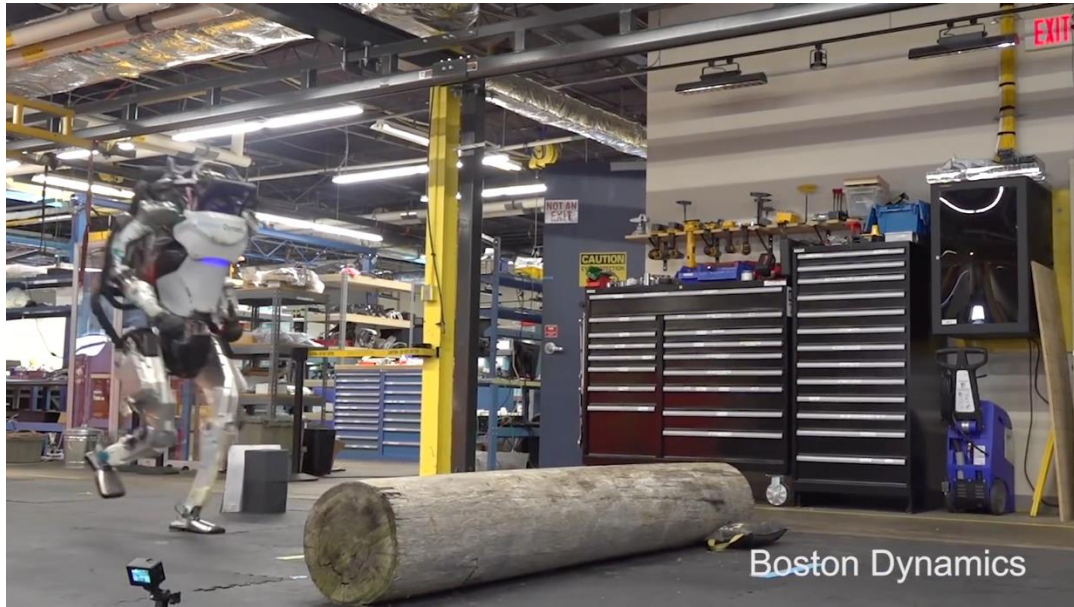
Why Do We Need Robot Learning?

- ❑ Data increases the “upper bound” of algorithm, computation, and hardware
 - Modeling the world could be hard
 - Environments/tasks might change
 - The policy structure might be limited
 - The optimization solver might be wrong
 - Assumptions might not hold
- ❑ Non-learning methods often separately design each module



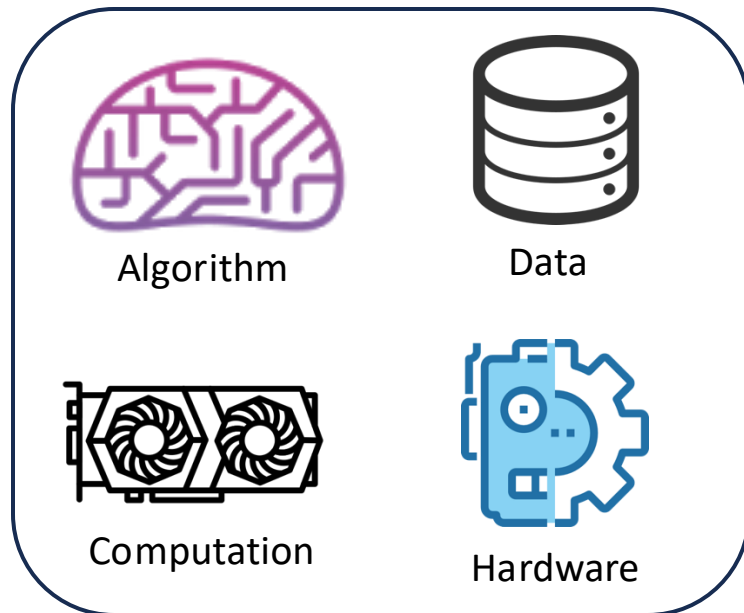
Why Do We Need Robot Learning?

- ❑ Data increases the “upper bound” of algorithm, computation, and hardware



A Glimpse of Today's Robot Learning

- ❑ We have learned all of them!
 - Model-free reinforcement learning (RL)
 - Sim2Real and Real2Sim2Real
 - Online adaptation
 - Imitation learning with demonstrations
 - Model-based RL
 - Offline RL



How far are we?

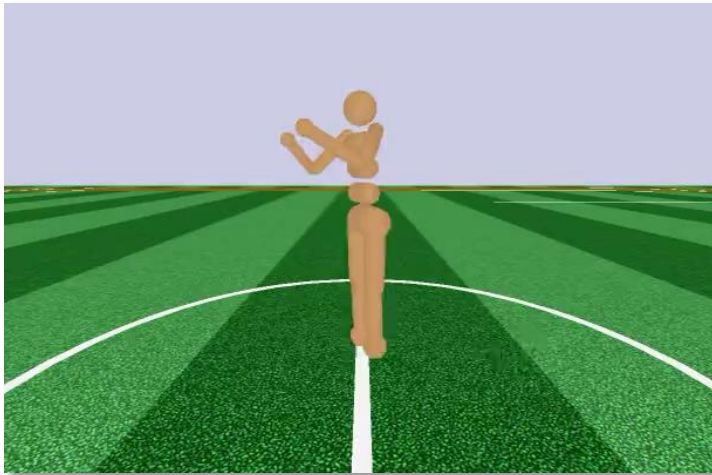


Embodied AGI in the real physical world



Where are We Today: Robot Learning in Simulation

❑ Robot in simulations



[Schulman *et al.*, 2015, 2017]



DM Control Soccer



Article | Published: 09 February 2022

Outracing champion Gran Turismo drivers with deep reinforcement learning

[Peter R. Wurman](#) , [Samuel Barrett](#), [Kenta Kawamoto](#), [James MacGlashan](#), [Kaushik Subramanian](#), [Thomas J. Walsh](#), [Roberto Capobianco](#), [Alisa Devlic](#), [Franziska Eckert](#), [Florian Fuchs](#), [Leilani Gilpin](#), [Piyush Khandelwal](#), [Varun Kompella](#), [HaoChih Lin](#), [Patrick MacAlpine](#), [Declan Oller](#), [Takuma Seno](#), [Craig Sherstan](#), [Michael D. Thomure](#), [Houmehr Aghabozorgi](#), [Leon Barrett](#), [Rory Douglas](#), [Dion Whitehead](#), [Peter Dürr](#), [Peter Stone](#), [Michael Spranger](#) & [Hiroaki Kitano](#) — [Show fewer authors](#)

Where are We Today: Robot Learning in Simulation

- ❑ GT Sophy overtakes the human driver in a fair way (no collision or blocking)
- ❑ How? DRL + distributed training platform + complex reward design



Captured on PS4. Game progression required to access most vehicles.

Where are We Today: Robot Learning in Simulation

- ❑ Eureka: pen spinning
- ❑ How? LLM for reward design

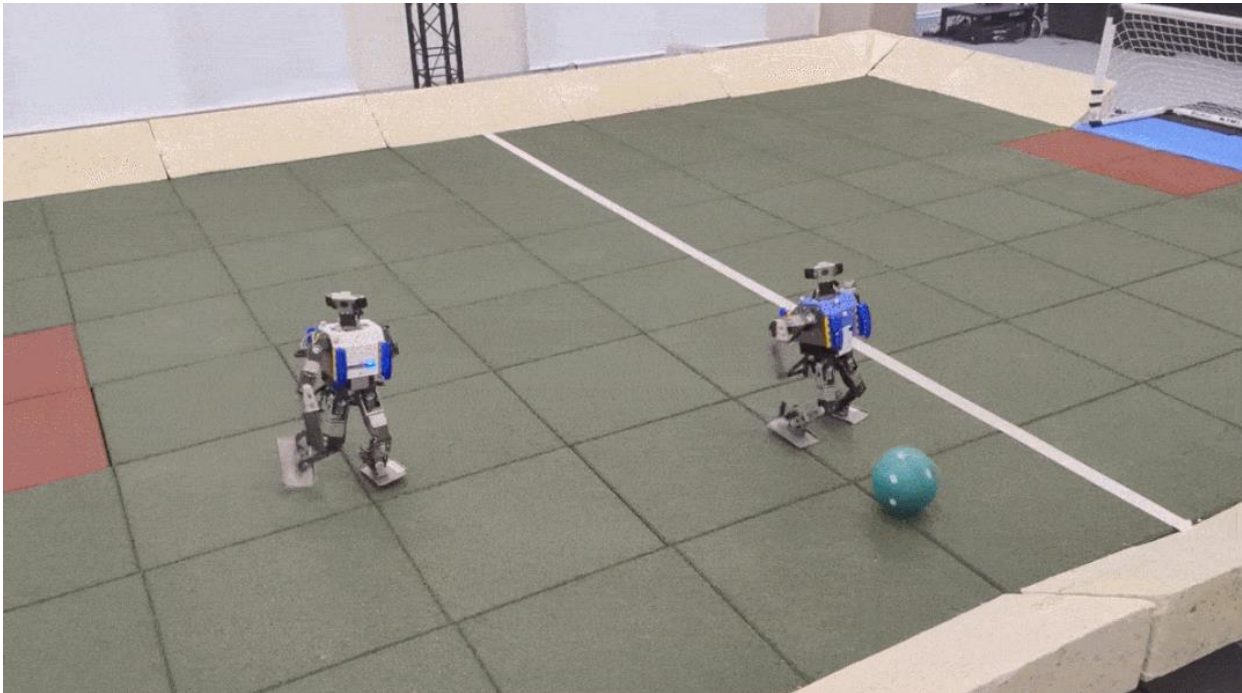
Eureka: Human-Level Reward Design via Coding Large Language Models

Jason Ma ^{1 2}, William Liang², Guanzhi Wang^{1 3}, De-An Huang¹,
Osbert Bastani², Dinesh Jayaraman², Yuke Zhu^{1 4}, Linxi "Jim" Fan ^{1 ‡}, Anima Anandkumar^{1 3 ‡}



Where are We Today: DRL + Sim2Real

- ❑ Train in simulation, deploy in the real world



1-1 Soccer (DeepMind)



2023-4-27

Learning Agile Soccer Skills for a Bipedal Robot with Deep Reinforcement Learning

Tuomas Haarnoja^{*1}, Ben Moran^{*1}, Guy Lever^{*1}, Sandy H. Huang^{*1}, Dhruva Tirumala¹, Markus Wulfmeier¹, Jan Humplik¹, Saran Tunyasuvunakool¹, Noah Y. Siegel¹, Roland Hafner¹, Michael Bloesch¹, Kristian Hartikainen^{2,4}, Arunkumar Byravan¹, Leonard Hasenclever¹, Yuval Tassa¹, Fereshteh Sadeghi^{3,4}, Nathan Batchelor¹, Federico Casarini¹, Stefano Saliceti¹, Charles Game¹, Neil Sreendra, Kushal Patel, Marlon Gwira, Andrea Huber¹, Nicole Hurley¹, Francesco Nori¹, Raia Hadsell¹ and Nicolas Heess¹

^{*}Equal contributions, ¹DeepMind, ²University of Oxford, ³Google, ⁴Work done at DeepMind

Where are We Today: DRL + Sim2Real

- ❑ Train in simulation, deploy in the real world



Tairan He*, Zhengyi Luo*, Wenli Xiao, Chong Zhang, Kris Kitani, Changliu Liu, Guanya Shi

Learning Human-to-Humanoid Real-Time Whole-Body Teleoperation (<https://human2humanoid.com/>)

Where are We Today: Learning from Demonstrations

- ❑ Collect human demonstrations and do imitation learning

Mobile ALOHA

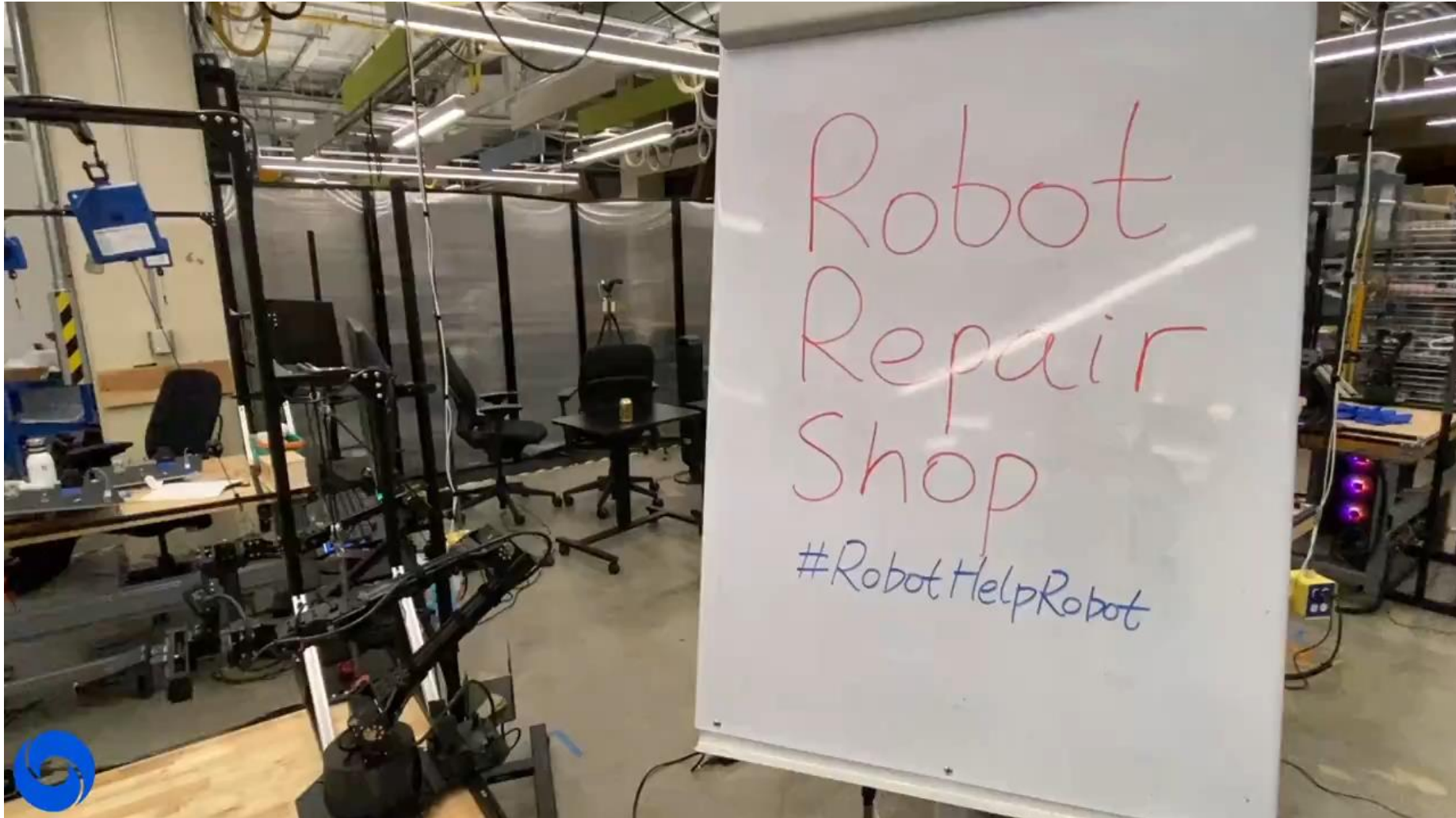
Learning Bimanual Mobile Manipulation with
Low-Cost Whole-Body Teleoperation

Zipeng Fu* Tony Z. Zhao* Chelsea Finn



Where are We Today: Learning from Demonstrations

- ❑ Scale up ALOHA in Google DeepMind



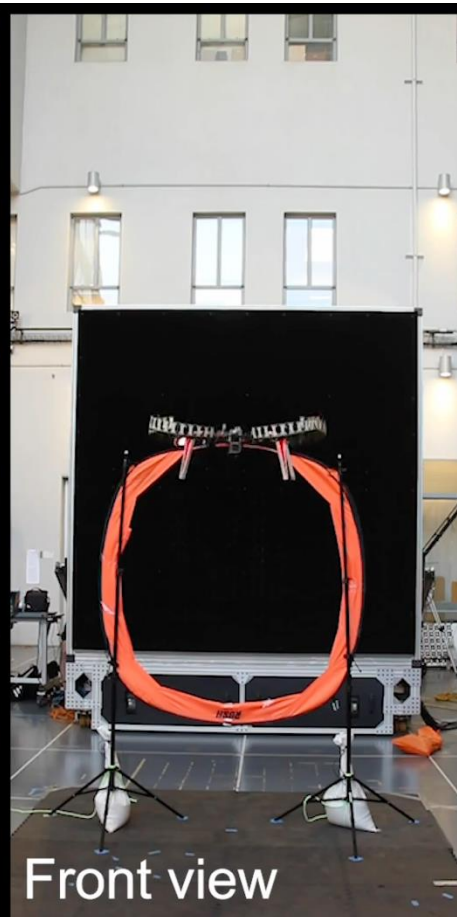
Where are We Today: Model-based RL

- ❑ No simulator. Collect data from real -> learn a model -> design a policy -> deploy
- ❑ Meta-learned dynamics model + online adaptive control



Side view

Our method can also generalize to the challenging time-varying wind condition.



Front view

Neural-Fly Enables Rapid Learning for Agile Flight in Strong Winds

Michael O'Connell[†], Guanya Shi[†], Xichen Shi,
Kamyar Azizzadenesheli, Anima Anandkumar, Yisong Yue, Soon-Jo Chung*

Division of Engineering and Applied Science, California Institute of Technology

[†] The first two authors contributed equally to this article. Alphabetical order.

* Corresponding authors. Email: sjchung@caltech.edu

This is the accepted version of *Science Robotics* Vol. 7, Issue 66, eabm6597 (2022)

DOI: 10.1126/scirobotics.abm6597 Video: <https://youtu.be/TuF9teCZX0U>

Data and training code: <https://github.com/aerorobotics/neural-fly>

Where are We Today: Model-based RL

- ❑ No simulator. Collect data from real -> learn a model -> design a policy -> deploy
- ❑ DNN dynamics + sampling-based MPC (MPPI)

Information Theoretic MPC for Model-Based Reinforcement Learning

Grady Williams, Nolan Wagener, Brian Goldfain, Paul Drews,
James M. Rehg, Byron Boots, and Evangelos A. Theodorou

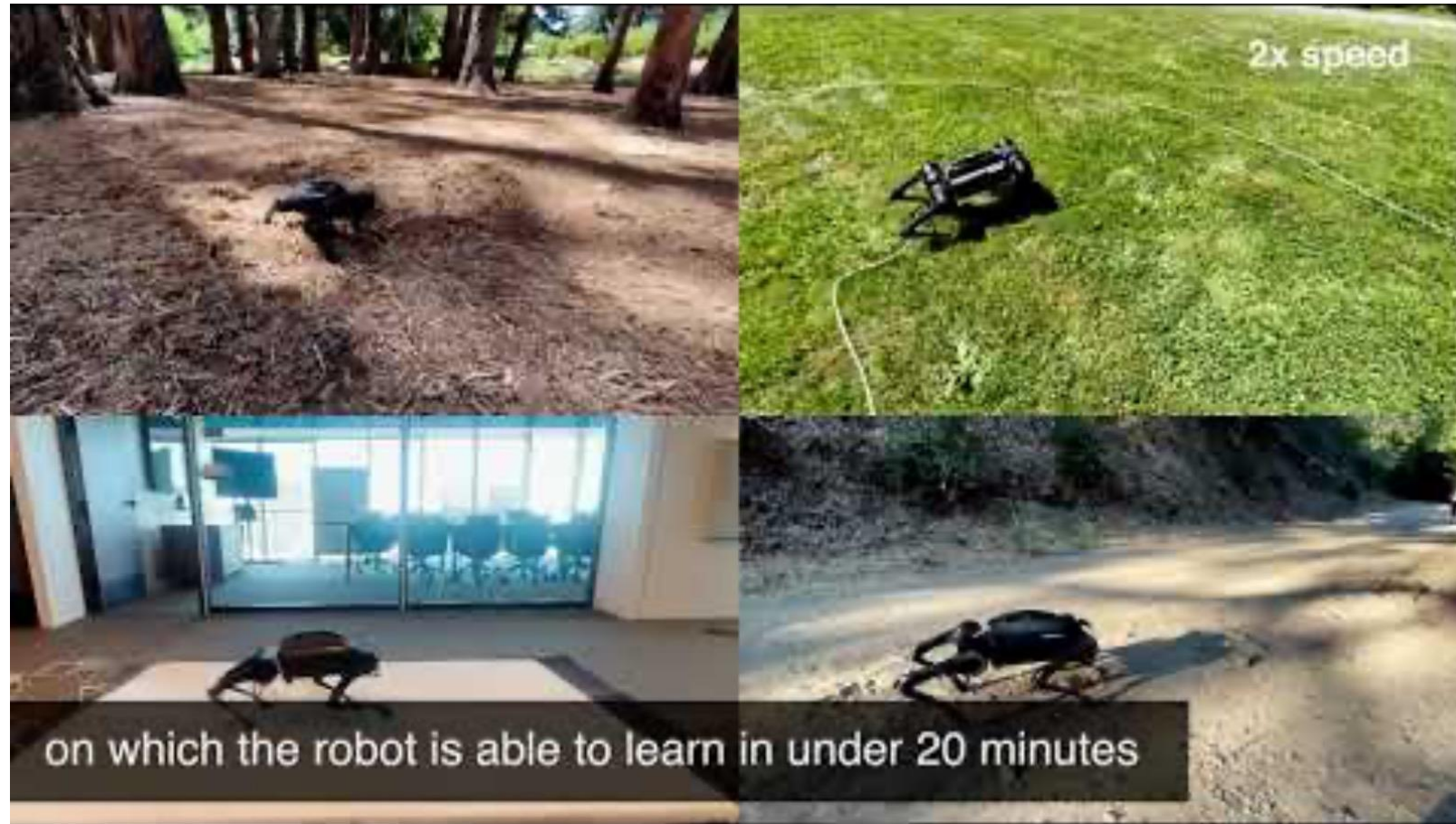


Where are We Today: Online Mode-free RL

- ❑ No simulator. MFRL from scratch

A Walk in the Park: Learning to Walk in 20 Minutes With Model-Free Reinforcement Learning

Laura Smith^{*1}, Ilya Kostrikov^{*1}, Sergey Levine¹
^{*}Equal contribution ¹Berkeley AI Research, UC Berkeley



Where are We Today: Offline RL

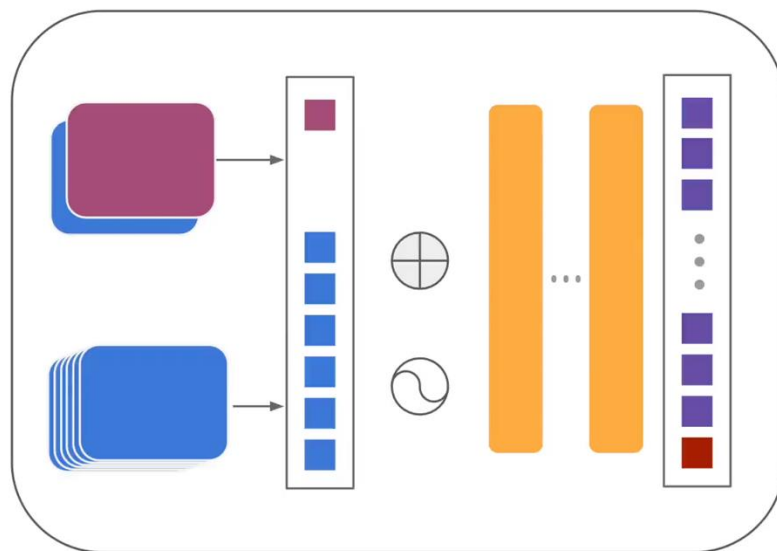
- ❑ No simulator. Learning from offline data.
- ❑ Difference from imitation learning: doesn't rely on expert data

General Navigation Models

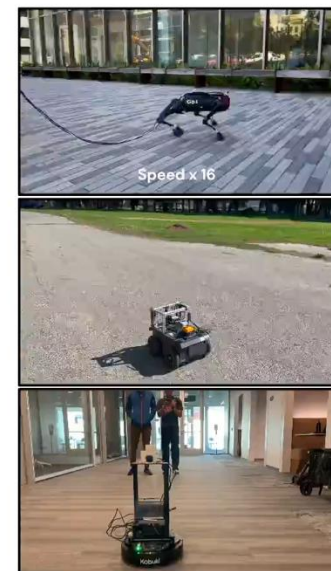
Dhruv Shah, Ajay Sridhar, Nitish Dashora, Catherine Glossop,
Kyle Stachowicz, Arjun Bhorkar, Kevin Black, Noriaki Hirose, Sergey Levine



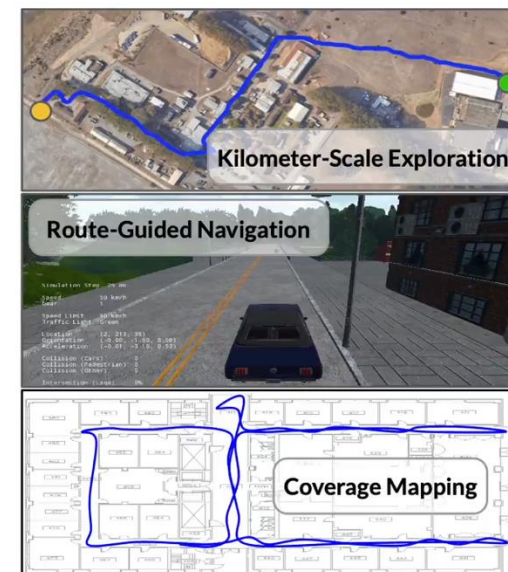
Diverse Training Data



General Navigation Models



Deploy Zero-Shot



Adapt to Downstream Tasks

Still Many Failures...

- ❑ These tasks seem “very simple” for humans



Still Many Failures...

- ❑ These tasks seem “very simple” for humans
- ❑ DARPA challenge 2015:



Moravec's Paradox

"The hard problems are easy and the easy problems are hard."

What to move where



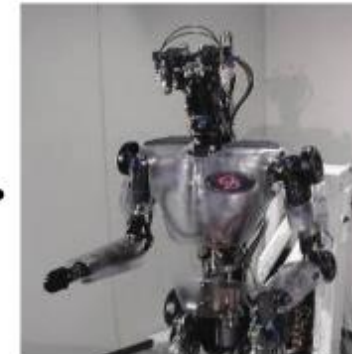
VS.



Moving



VS.



Moravec's Paradox



Hans P. Moravec
Principal Research Scientist
Robotics Institute
Carnegie Mellon University

“It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to *perception* and *mobility*.” (1988)

Common Belief



Starting of Life

Apes

Humans

Language



3.5B Years

~18M Years

~50K-500K Years

Today's Agenda

Today is going to be a “causal” lecture! Let's make it interactive so raise your hand for any questions at anytime.

- ☐ Rethink: What is robot learning and why is it so hard?
- ☒ (Briefly) review what we have learned
- ☐ Open challenges and discussions
- ☐ Q&A! Ask me anything!

Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

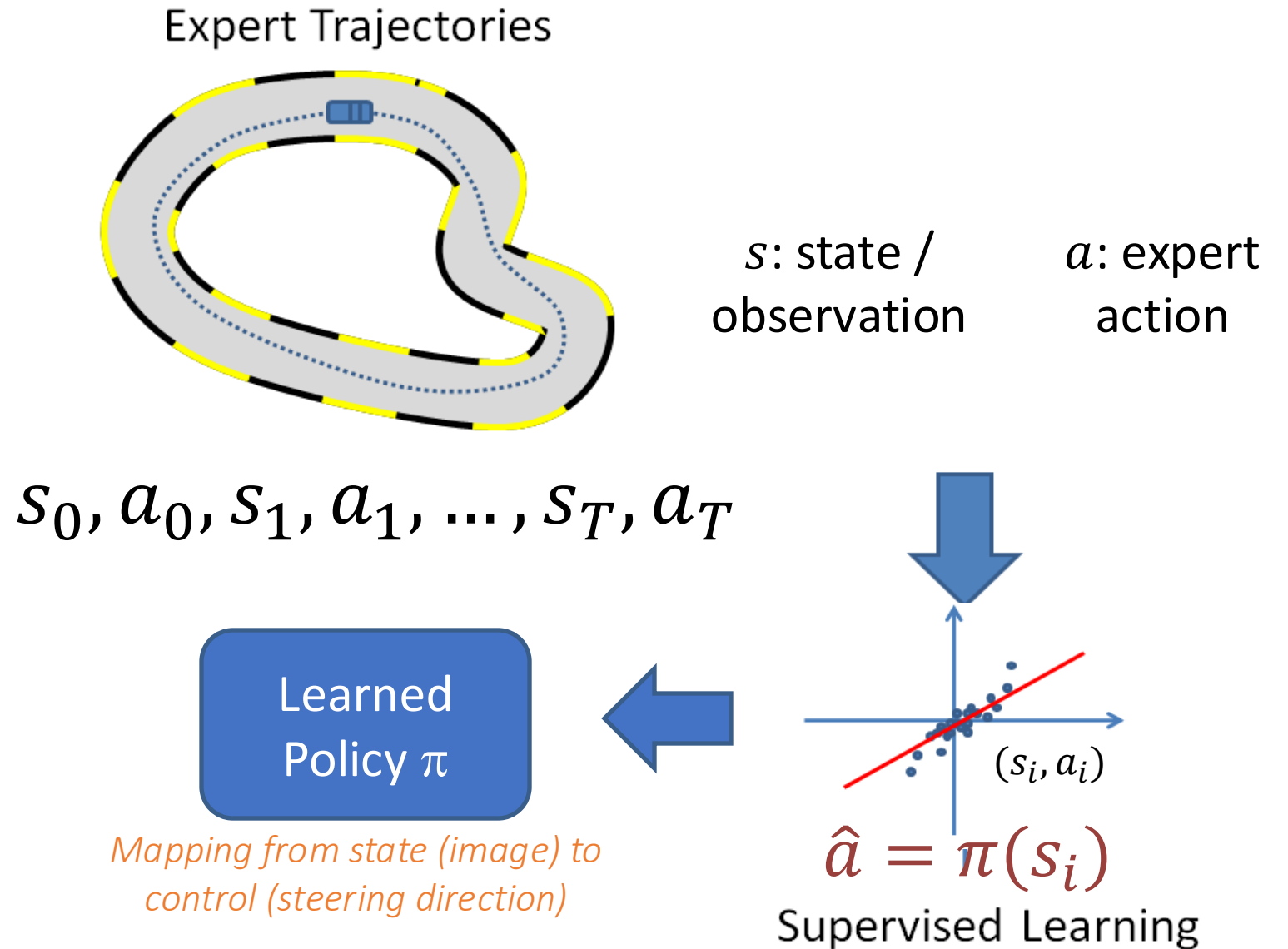
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☒ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

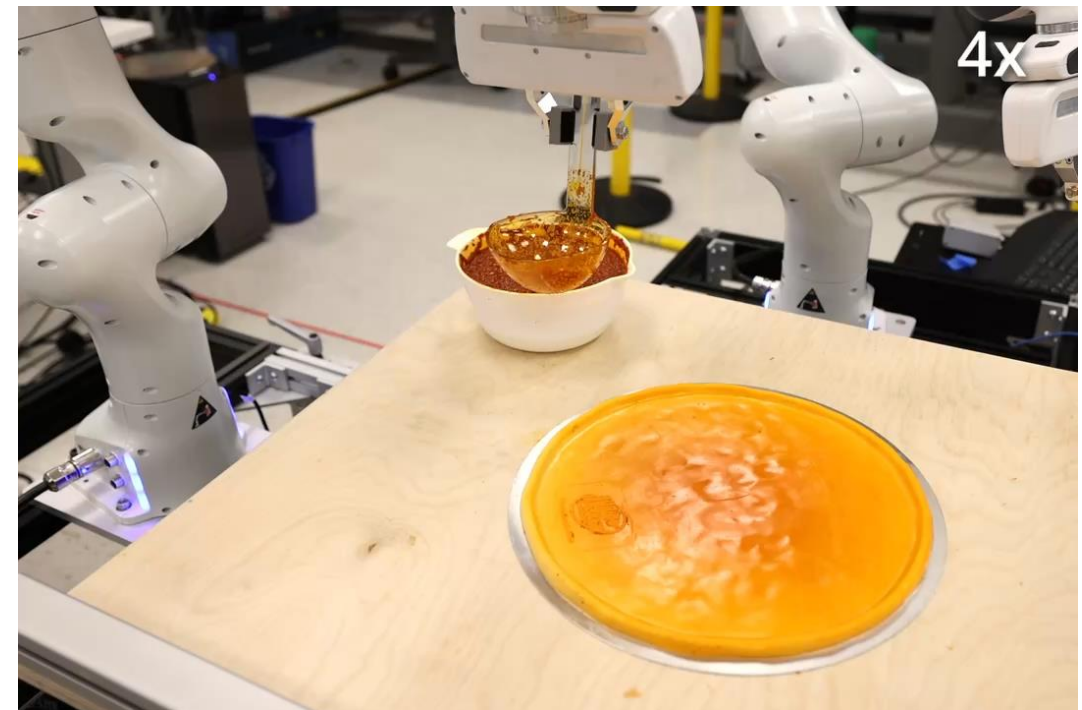
Imitation Learning

- ❑ How is learning from demonstrations different from standard supervised learning?
- ❑ How to design efficient and effective algorithms?



Imitation Learning

- ❑ You will learn how these demos work
 - Left: Mobile ALOHA; Right: Diffusion policy.



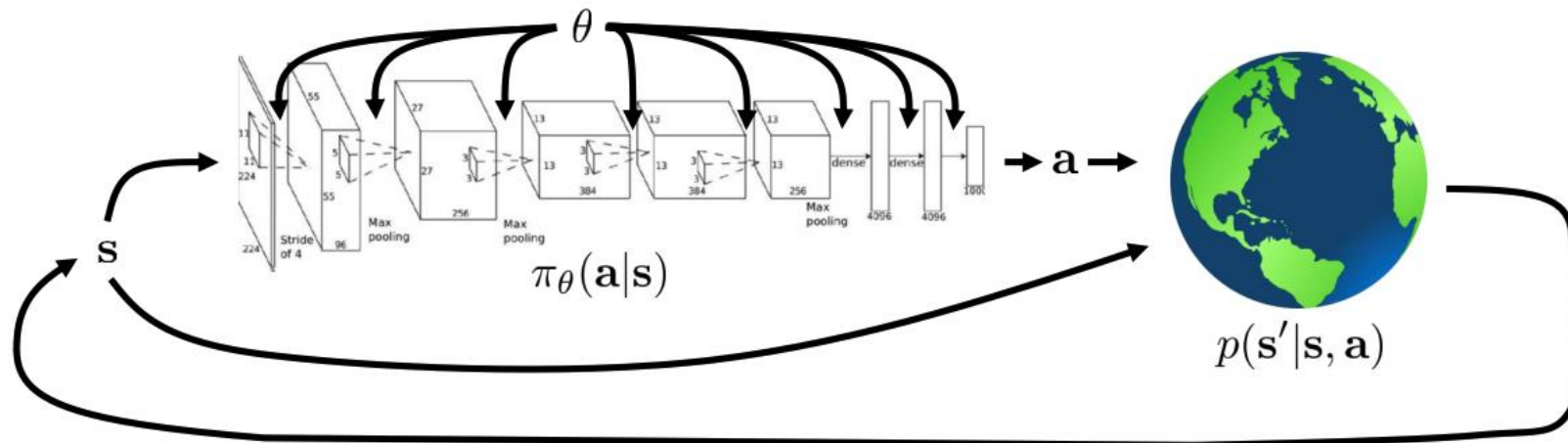
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☒ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Model-free Reinforcement Learning

- ❑ Learning from directly interacting with the environment
- ❑ Learning from trial-and-errors

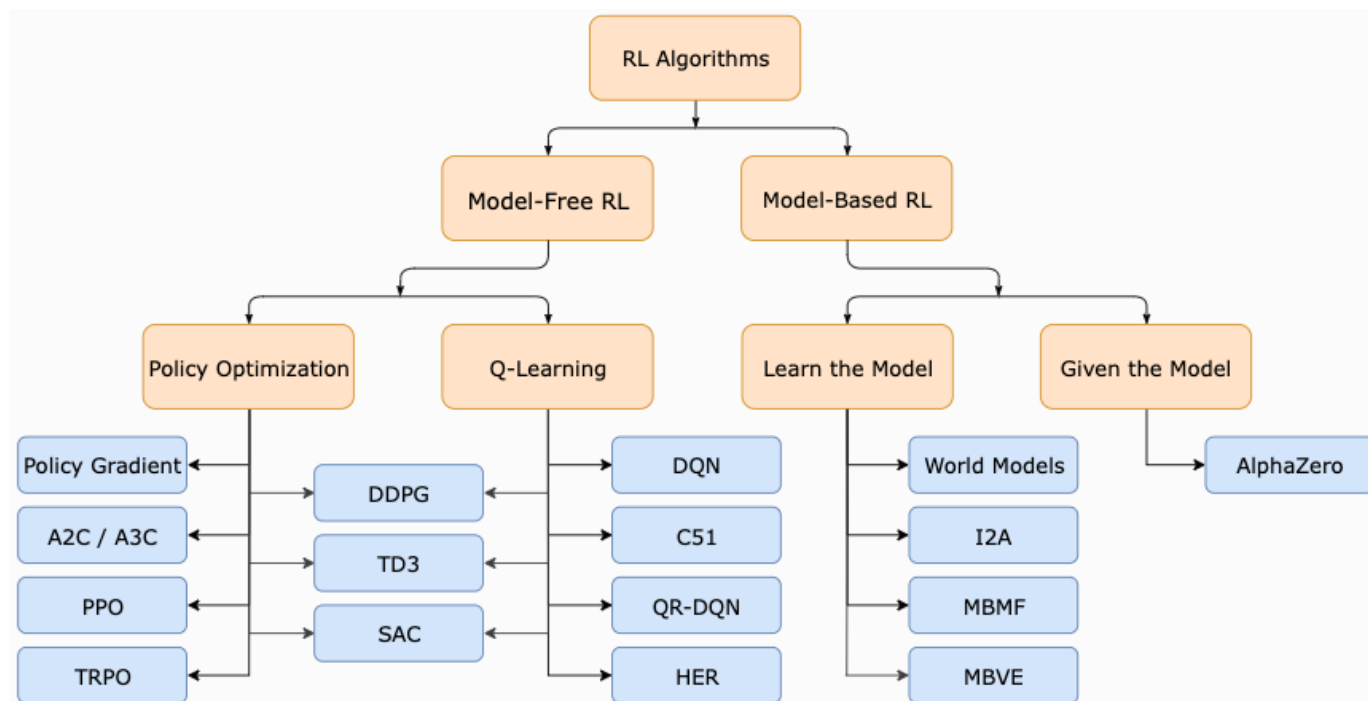
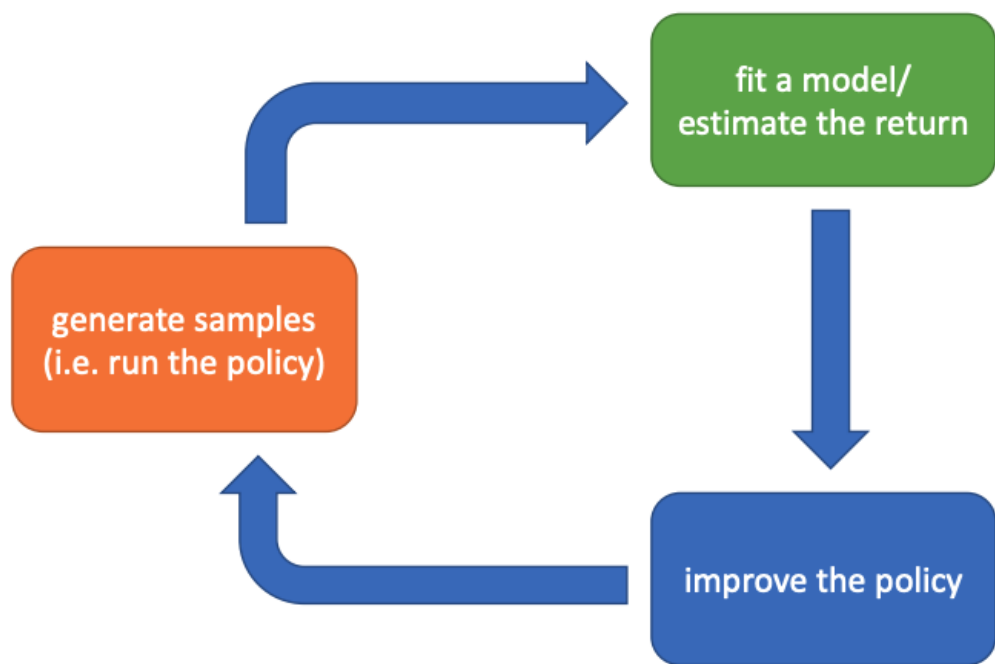


$$\underbrace{p_{\theta}(s_1, a_1, \dots, s_T, a_T)}_{\pi_{\theta}(\tau)} = p(s_1) \prod_{t=1}^T \pi_{\theta}(a_t | s_t) p(s_{t+1} | s_t, a_t)$$

$$\theta^* = \arg \max_{\theta} E_{\tau \sim p_{\theta}(\tau)} \left[\sum_t r(s_t, a_t) \right]$$

Taxonomy of MFRL Algorithms

- ❑ **Value-based** (example: FQI, DQN): estimate value function or Q-function of the optimal policy (no explicit policy).
- ❑ **Policy-gradient** (example: REINFORCE, NPG): directly optimize the policy
- ❑ **Actor-critic** (example: SAC, DDPG): estimate value function or Q-function of the current policy, use it to improve policy



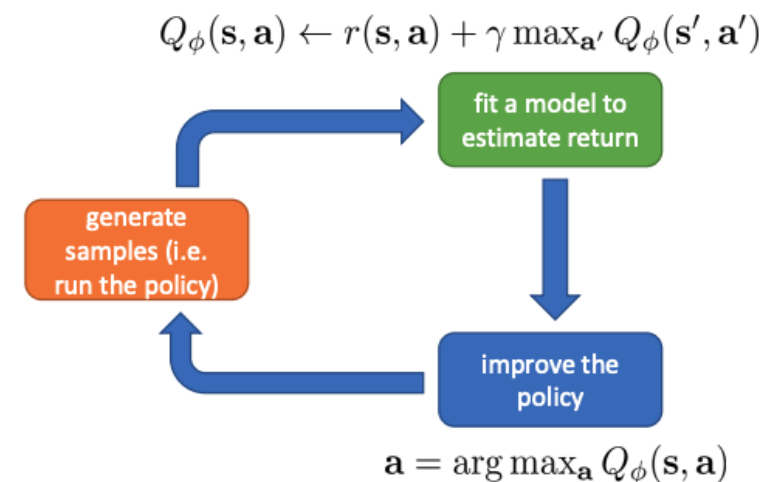
Recap: Q-Learning

□ Based on: **Bellman optimality equation**

$$Q^*(s, a) = \sum_{s', r} p(s', r | s, a) [r + \gamma \cdot \max_{a'} Q^*(s', a')]$$

□ Example: DQN (Deep Q-Network):

- Epsilon greedy policy gives a and observe (s, a, s', r)
- Update the replay buffer
- Sample a batch from the buffer. Compute the target $y_i = r(s_i, a_i) + \gamma \max_{a'_i} Q_{\phi^-}(s'_i, a'_i)$
- Update: $\phi \leftarrow \phi - \eta \sum_i (Q_{\phi}(s_i, a_i) - y_i) \nabla_{\phi} Q_{\phi}(s_i, a_i)$
- Periodically update ϕ^- : $\phi^- \leftarrow \phi$



Recap: Policy Gradient

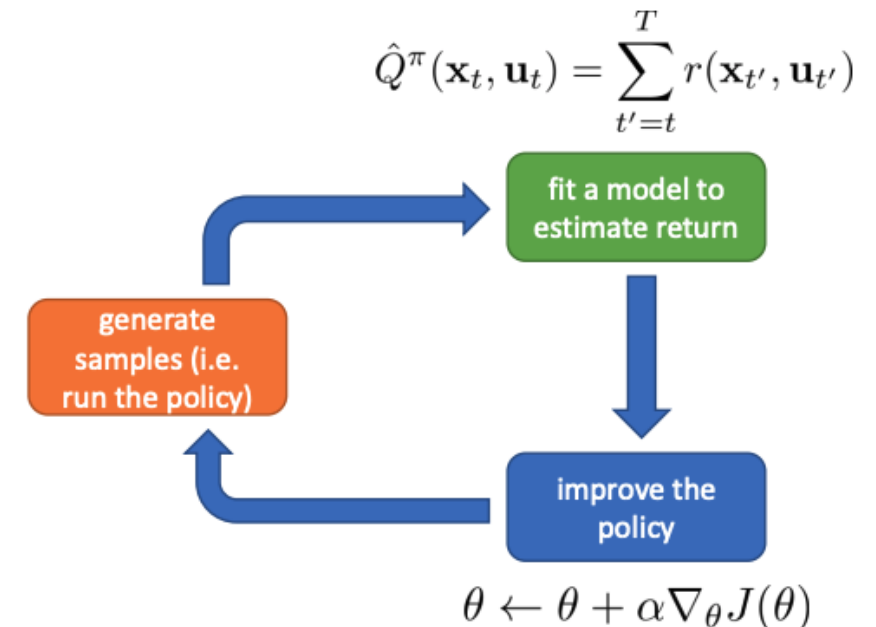
□ Based on: **Policy gradient theorem**

$$\underbrace{p_{\theta}(\mathbf{s}_1, \mathbf{a}_1, \dots, \mathbf{s}_T, \mathbf{a}_T)}_{p_{\theta}(\tau)} = p(\mathbf{s}_1) \prod_{t=1}^T \underbrace{\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t) p(\mathbf{s}_{t+1} | \mathbf{s}_t, \mathbf{a}_t)}_{J(\theta)}$$
$$\theta^* = \arg \max_{\theta} E_{\tau \sim p_{\theta}(\tau)} \left[\sum_t r(\mathbf{s}_t, \mathbf{a}_t) \right]$$

$$\nabla_{\theta} J(\theta) = E_{\tau \sim p_{\theta}(\tau)} [R(\tau) \sum_t \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)]$$

□ Example: REINFORCE

- 
1. sample $\{\tau^i\}$ from $\pi_{\theta}(\mathbf{a}_t | \mathbf{s}_t)$ (run the policy)
 2. $\nabla_{\theta} J(\theta) \approx \sum_i (\sum_t \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}_t^i | \mathbf{s}_t^i)) (\sum_t r(\mathbf{s}_t^i, \mathbf{a}_t^i))$
 3. $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$



Recap: Actor-Critic

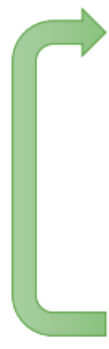
□ Based on: **Policy gradient with baselines**

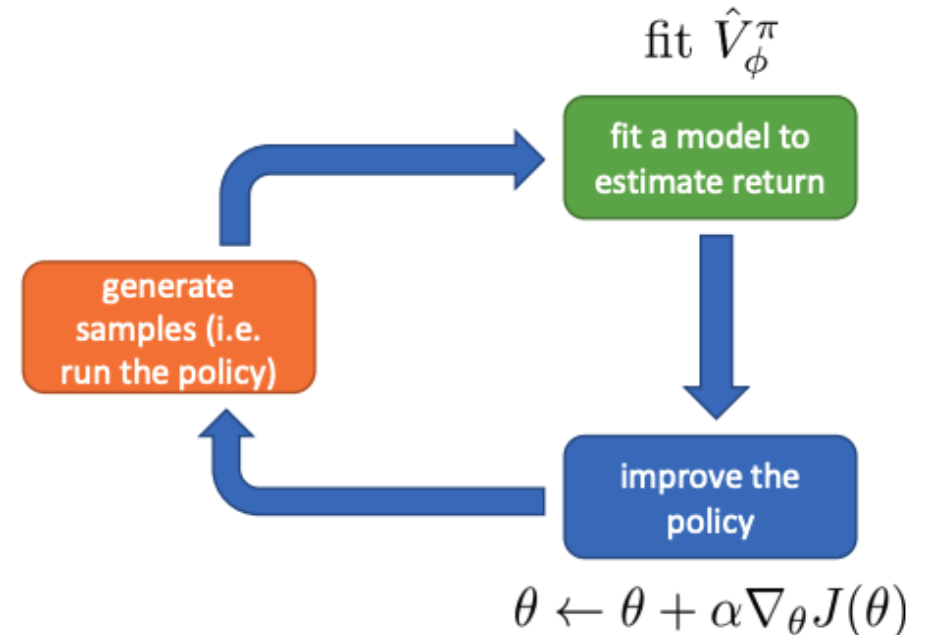
$$\nabla_{\theta} J(\theta) \propto E_{s \sim p_{\theta}, a \sim \pi_{\theta}} [(Q^{\pi}(s, a) - V^{\pi}(s)) \nabla_{\theta} \log \pi_{\theta}(a|s)]$$

- Learn $V_{\phi}^{\pi}(s)$
- Estimate $A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s)$ by $r + \gamma V_{\phi}^{\pi}(s') - V_{\phi}^{\pi}(s)$

□ Example: online actor-critic

online actor-critic algorithm:

- 
1. take action $\mathbf{a} \sim \pi_{\theta}(\mathbf{a}|\mathbf{s})$, get $(\mathbf{s}, \mathbf{a}, \mathbf{s}', r)$
 2. update $\hat{V}_{\phi}^{\pi}$ using target $r + \gamma \hat{V}_{\phi}^{\pi}(\mathbf{s}')$
 3. evaluate $\hat{A}^{\pi}(\mathbf{s}, \mathbf{a}) = r(\mathbf{s}, \mathbf{a}) + \gamma \hat{V}_{\phi}^{\pi}(\mathbf{s}') - \hat{V}_{\phi}^{\pi}(\mathbf{s})$
 4. $\nabla_{\theta} J(\theta) \approx \nabla_{\theta} \log \pi_{\theta}(\mathbf{a}|\mathbf{s}) \hat{A}^{\pi}(\mathbf{s}, \mathbf{a})$
 5. $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$



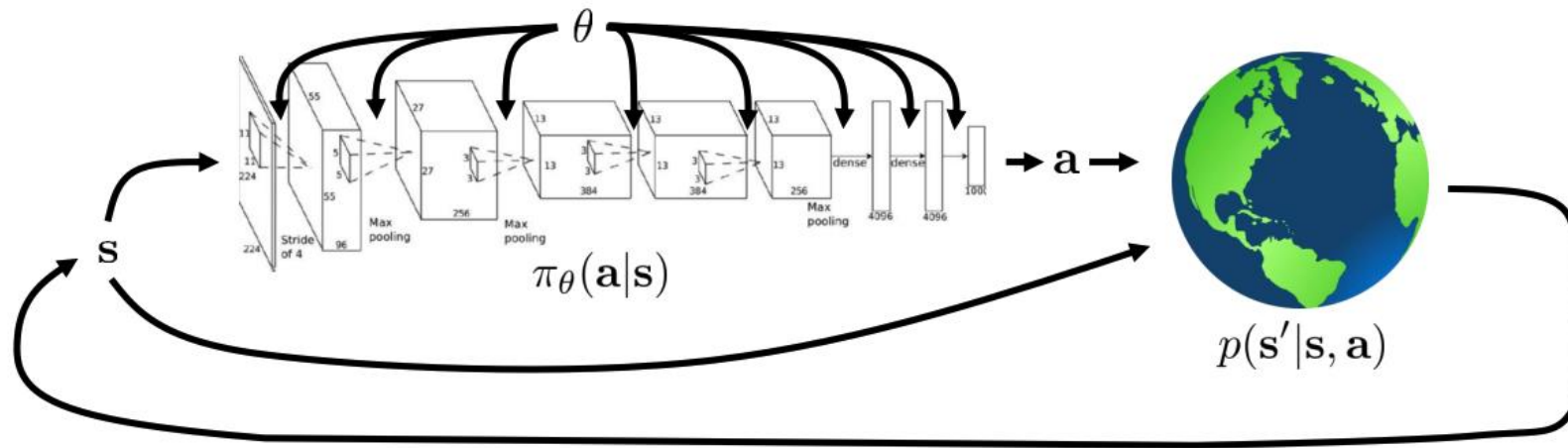
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☒ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Model-based Control and RL

- Build/learn a model (transition dynamics, “world model”) and then control



$$\underbrace{p_\theta(s_1, a_1, \dots, s_T, a_T)}_{\pi_\theta(\tau)} = p(s_1) \prod_{t=1}^T \pi_\theta(a_t | s_t) p(s_{t+1} | s_t, a_t)$$

$$\theta^* = \arg \max_{\theta} E_{\tau \sim p_\theta(\tau)} \left[\sum_t r(s_t, a_t) \right]$$

model-free RL assumes it is unknown
(don't even attempt to learn it)

Model-based Control and RL

□ We have learned:

- Model-based control (with known model): PID, Lyapunov, etc.
- Optimal control (with known model): LQR, iLQR, DDP, SQP, etc.
- Model-based reinforcement learning (need to learn dynamics): CEM, MPPI, Neural-Control, etc.
- Deep MBRL: Dreamer, TD-MPC, etc.

$$\begin{array}{ll} \min_{x,u} & \sum_{t=1}^T c_t(x_t, u_t) \\ \text{s. t.} & \left\{ \begin{array}{l} x_{t+1} = f_t(x_t, u_t) \\ x_t \in X_t \\ u_t \in U_t \end{array} \right. \end{array}$$

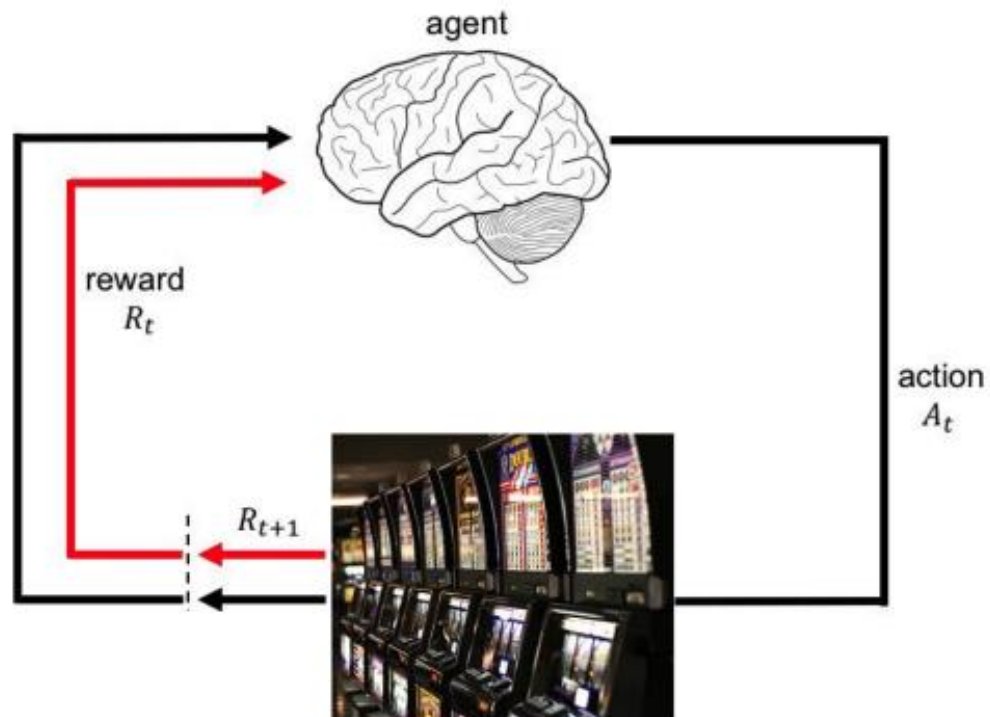
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☒ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Bandit and Exploration

- ❑ Bandit is the simplest RL problem (no state)
 - Therefore, it allows us to more fundamentally reason exploration v.s. exploitation
 - Many algorithmic principles from bandits can be transferred to RL
- ❑ We also learned preference-based learning (no reward, only relative preference)



Explain reinforcement learning to a 6 year old.

A

In reinforcement learning, the agent is...

B

Explain rewards...

C

In machine learning...

D

We give treats and punishments to teach...

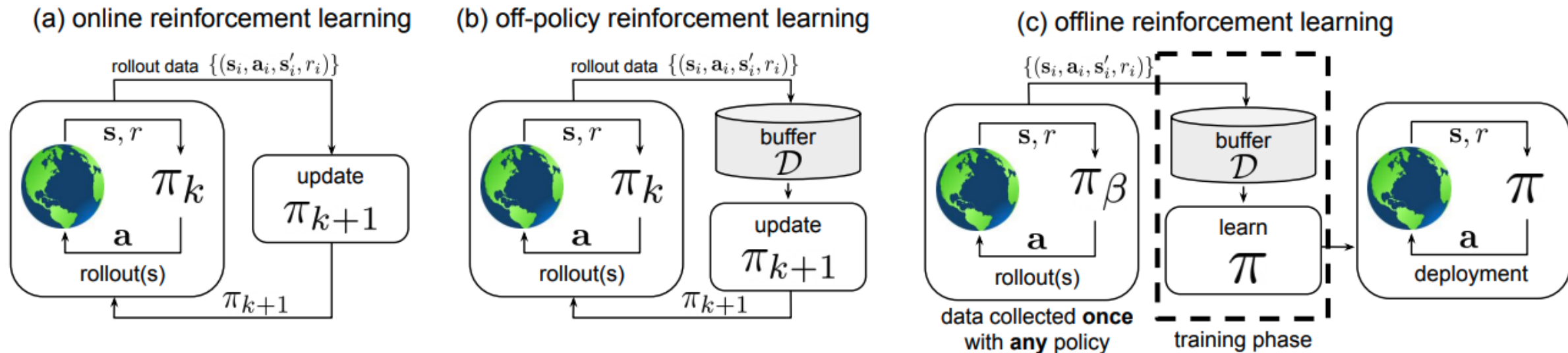
Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☒ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☐ Topic group 8: Specialized topics (one guest lecture)
- ☐ Project presentation

Reinforcement Learning from Offline Data

- ❑ Motivation: Online RL relies exclusively on interaction
 - Expensive and unsafe
- ❑ RL from offline dataset? Question:
 - How to learn?
 - How is it different from imitation learning?
- ❑ We also learned inverse RL



Course Structure

Latest schedule: <https://16-831.github.io/fall24/lectures>

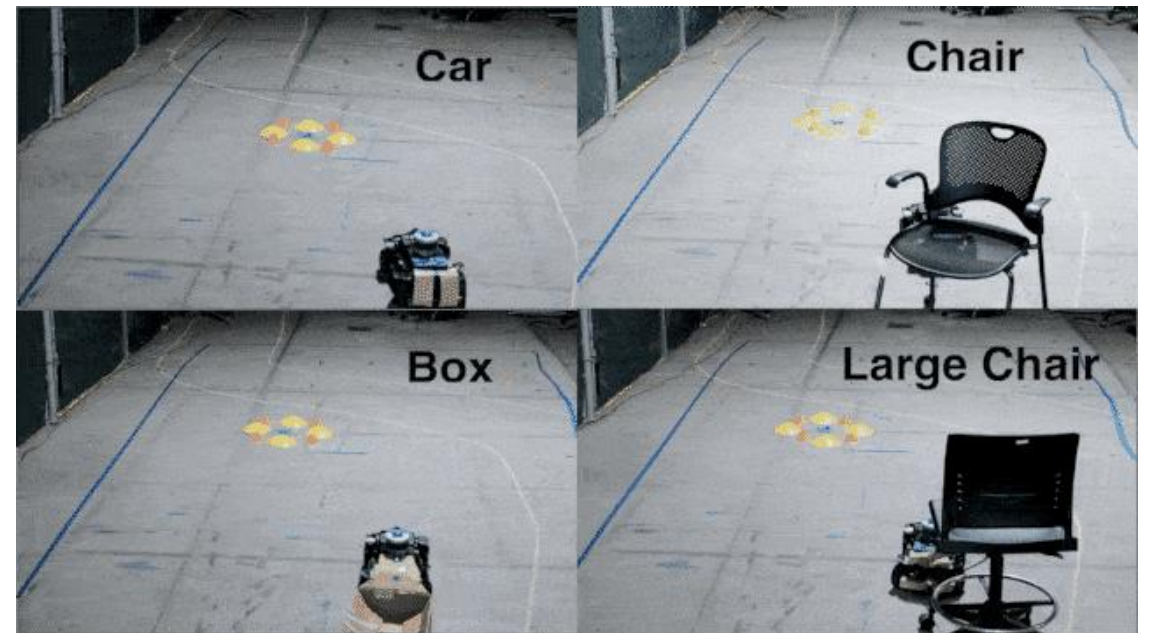
- ☐ Topic group 1: Course introduction: What is robot learning and why?
- ☐ Topic group 2: Machine/Deep learning refresher
- ☐ Topic group 3: Imitation learning
- ☐ Topic group 4: Model-free reinforcement learning
- ☐ Topic group 5: Model-based reinforcement learning
- ☐ Topic group 6: Bandit, preference-based learning, exploration
- ☐ Topic group 7: Reinforcement learning from offline data (one guest lecture)
- ☒ Topic group 8: Specialized topics (**one guest lecture**)
- ☐ Project presentation

Specialized Topics

- ❑ Topic 1: Safe RL and safe robot learning
 - Collision-free, smooth, stable, etc.
- ❑ Topic 2: Multi-task/adaptive/transferrable robot learning
 - Adapt to different dynamics (e.g., different robots)
 - Adapt to different tasks (e.g., goal-conditioned RL)



Wabersich et al., Data-Driven Safety Filter, 2023



Xiao* and He* et al., Safe Deep Policy Adaptation

Specialized Topics

- ❑ Topic 3: Robot simulator and sim2real
 - Introduce various simulators
 - Introduce various sim2real (and real2sim2real) methods

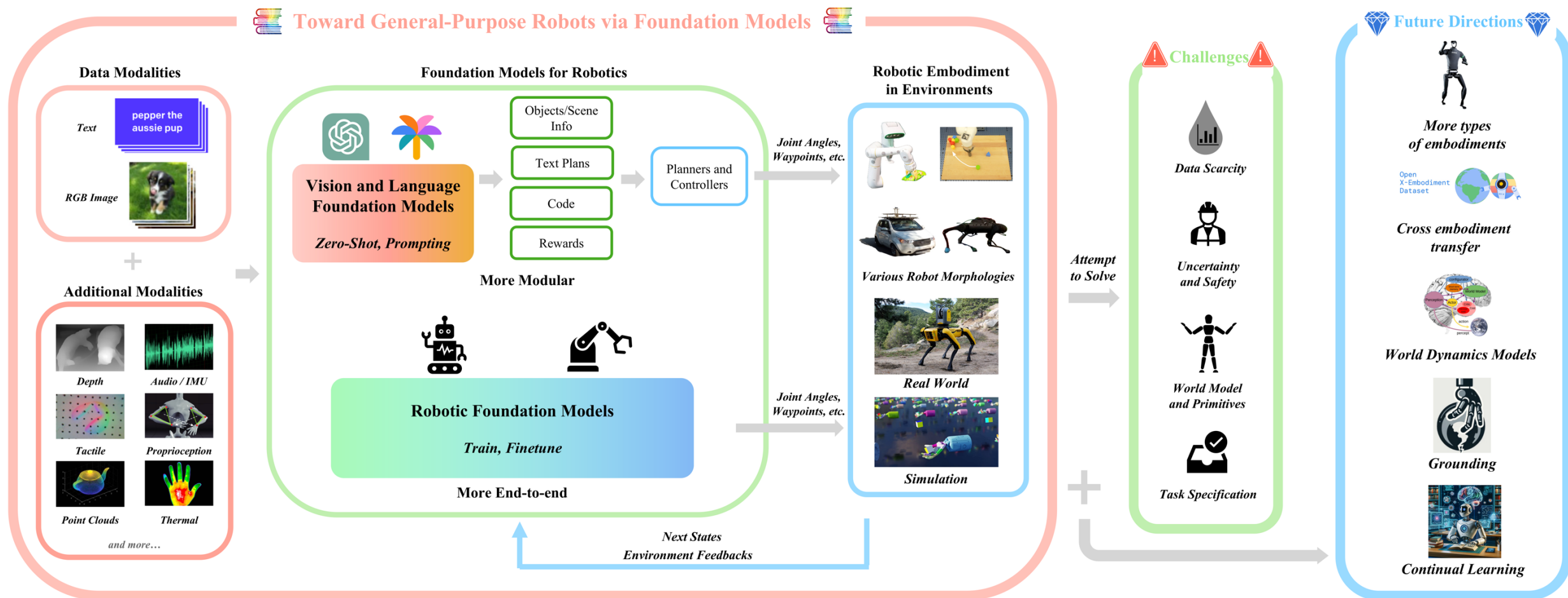


NVIDIA Isaac Gym



Specialized Topics

❑ Topic 4: Foundation models in robotics



Relation to Other CMU Courses

- ❑ 10-703 Deep Reinforcement Learning (Katerina Fragkiadaki)
 - Focus on generic sequential decision making
 - 16-831 focuses on robotics
 - Recommend if you want to go deeper in generic DRL

- ❑ 16-881 Seminar: Deep Reinforcement Learning for Robotics (Deepak Pathak)
- ❑ 16-884 Deep Learning for Robotics (David Held)
 - Pure paper discussions or mix of lectures and paper discussions
 - Not basic, assume prior background
 - Recommend if you want to learn latest robot learning research

Relation to Other CMU Courses

❑ 16-745 Optimal Control and Reinforcement Learning (Zachary Manchester)

- Focus on optimization-based control and model-based RL
- Recommend if you want to know more about lecture 13-14

2/26	Lecture 12: Model-Based Control Basics [slides]	[Feedback Systems Textbook]	HW2 Due; HW3 Out	● Model- Based RL
2/28	Lecture 13: Optimal Control and Planning Part 1 [slides]	[Murray's Notes][iLQR] [DDP][SCP]		● Model- Based RL
3/04	Spring Break			
3/06	Spring Break			
3/11	Lecture 14: Optimal Control and Planning Part 2 [slides]	[PETS][Neural-Control Family][MPPI][PILCO] [MBPO]		● Model- Based RL
3/13	Lecture 15: Deep Model-Based RL [slides]	[Dreamer][TD-MPC]	HW3 Due; HW4 Out	● Model- Based RL

❑ 16-899 Adaptive Control and Reinforcement Learning (Changliu Liu)

- Advanced course focusing on control theory, safe control, Lyapunov stability, etc
- Recommend if you want to know more about lecture 12 and safe RL

Today's Agenda

Today is going to be a “causal” lecture! Let's make it interactive so raise your hand for any questions at anytime.

- ☐ Rethink: What is robot learning and why is it so hard?
- ☐ (Briefly) review what we have learned
- ☒ Open challenges and discussions
- ☐ Q&A! Ask me anything!

What is “Robot Learning”?

- ❑ **Learning** to make **sequential** decisions in the **physical world**
- ❑ **Learning**: Data-driven and improve from data
 - W/o learning & data: search & planning, classic control, optimal control, ...
- ❑ **Sequential**: The current action/decision influences the next state therefore the next action/decision
 - W/o sequential: bandit, standard supervised learning, ...
- ❑ **Physical world**: The robot needs to interact with the physical world in the closed-loop
 - A.k.a. “embodied intelligence”
 - W/o physical world: RL for games, LLMs...

Uniqueness of “Robot Learning”

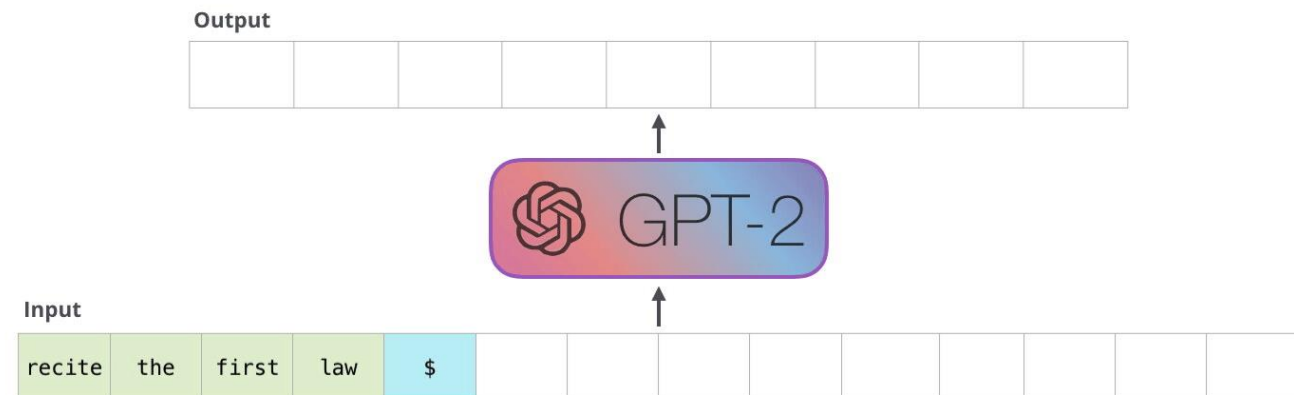
- ❑ **Learning**: Data-driven and improve from data
 - W/o learning & data: search & planning, optimization, control theory, ...

Challenges (compared to other AI or machine learning domains):

- ❑ Where is the data from?
- ❑ How to use the data?
- ❑ What does “robotics foundation model” mean, and how to train it?

Why do GPTs (and other VLMs/LLMs) work? How much does it apply to robot learning?

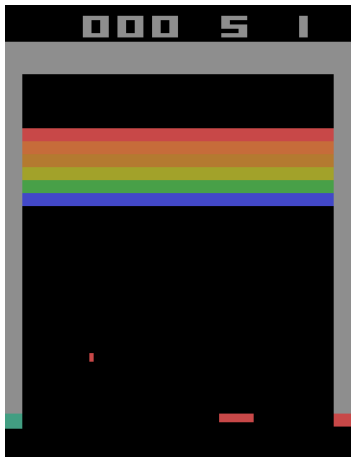
- ❑ Architecture: transformer
- ❑ Data: web texts, books, wikis, ...
- ❑ Loss: next token prediction
- ❑ Optimization: SGD
- ❑ Generation: autoregressive



Uniqueness of “Robot Learning”

- ❑ **Sequential:** The current action/decision influences the next state therefore the next action/decision
 - W/o sequential: bandit, standard supervised learning, ...
- ❑ **Physical world:** The robot needs to interact with the physical world in the closed-loop
 - A.k.a. “embodied intelligence”
 - W/o physical world: RL for game, ...

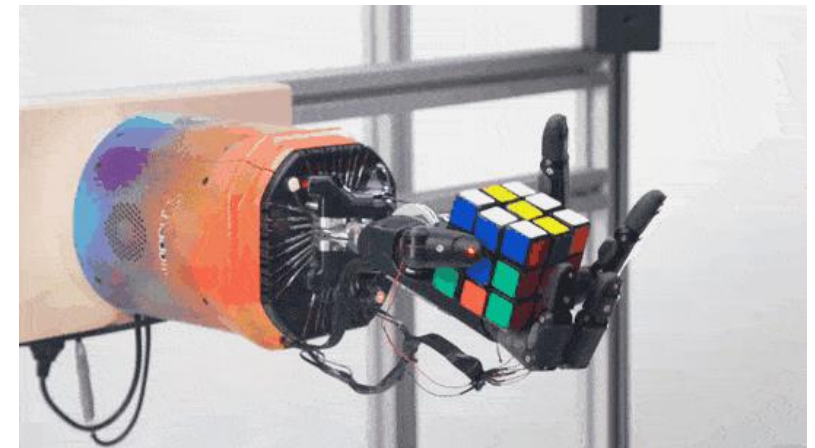
What are challenges (compared to DRL in games such as Alpha Go)?



DQN for Atari (DeepMind)



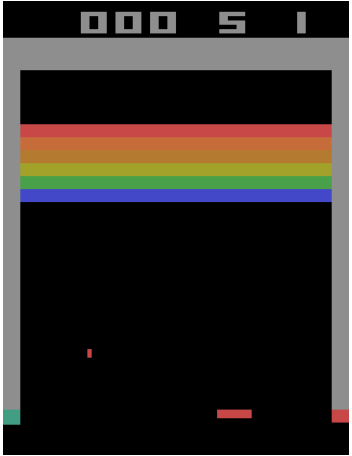
AlphaGo (DeepMind)



Manipulator (OpenAI)

Uniqueness of “Robot Learning”

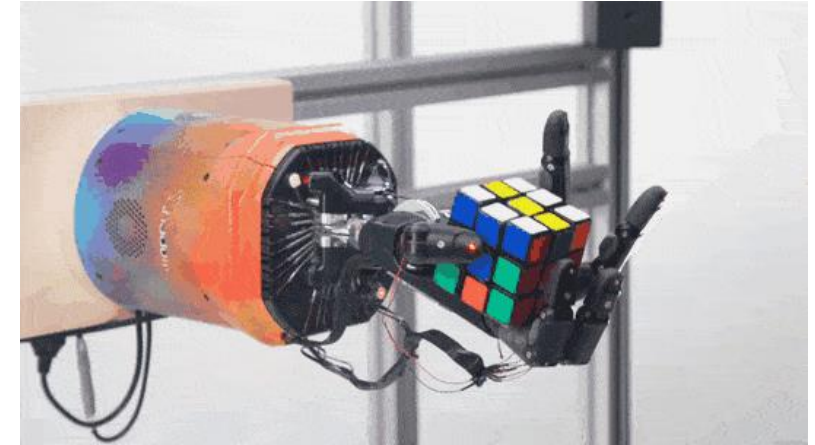
What are challenges (compared to DRL in games such as Alpha Go)?



DQN for Atari (DeepMind)



AlphaGo (DeepMind)



Manipulator (OpenAI)

- ☐ Known & static v.s. Unknown & dynamic
- ☐ One specific task v.s. Many tasks
- ☐ Clear optimization goals (i.e., reward) v.s. Unclear goals
- ☐ Offline learning is enough v.s. Require online adaptation
- ☐ Slow action v.s. Relatively fast real-time action (e.g., 50Hz)
- ☐ Allow failures v.s. Physics doesn't forgive!
- ☐ Discrete digital world v.s. Continuous physical world



Today's Agenda

Today is going to be a “causal” lecture! Let's make it interactive so raise your hand for any questions at anytime.

- ☐ Rethink: What is robot learning and why is it so hard?
- ☐ (Briefly) review what we have learned
- ☐ Open challenges and discussions
- ☒ Q&A! Ask me anything!

Thank You!

(And Don't Forget the Course Evaluation)

SmartEvals!



To ensure high FCE response rates, you can ask your class to scan this QR code into their cell phones and complete their ROB 16831 A taught by: SHI Lect Graduate 2024 Fall evaluations.

This QR code will require the student to type in their login id to the SmartEvals.com system but *not* their password.

QR codes are more secure than a typical paper based faculty course evaluation.
This code can be set to be used for 30 minutes after the first student scans the code.